Currículum Vitae

José Roberto Tello Ayala

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EDUCATION

Ph.D. Candidate, Applied Mathematics
Harvard University
Advised by Professor Finale Doshi-Velez at Harvard's Data to Actionable Knowledge Lab, my research focuses on Machine Learning model interpretability, particularly in computer vision applications for healthcare. I am affiliated with the Cardiovascular Disease Initiative (CVDi) and the Eric and Wendy Schmidt Center at the Broad Institute of Harvard and MIT and the Cardiovascular Research Center (CVRC) at Massachusetts General Hospital (MGH).
Ongoing since August 31 of 2022. Expected graduation: May 25 2027

Bachelor's degree in Science, Applied Mathematics
Instituto Tecnológico Autónomo de México (ITAM)
Graduated with honors by acquiring a special mention thanks to my thesis titled: "A study concerning the Chebyshev Center problem in finite-dimensional normed vector spaces" under the supervision of Dr. César García-García.
From August of 2015 to December of 2019.

PUBLICATIONS

CorAI: Artificial Intelligence Based Detection of Coronary Artery Stenosis in more than 40,000 Coronary Angiograms
Authors: Akl C. Fahed*, **Jose Roberto Tello Ayala***, Samuel Friedman*, Ozan Unlu, Marcus D. R. Klarqvist, Paolo Di Achille, Mustafa Nasir-Moin, Eugene Pomerantsev, Mahnaz Maddah, Finale Doshi-Velez, Patrick T. Ellinor, Anthony Philippakis
In preparation.

An Information-theoretic Lens on Saliency Map Methods
Authors: **Jose Roberto Tello Ayala**, Weiwei Pan, Siddharth Swaroop, Akl C. Fahed, Eugene V. Pomerantsev, Patrick Thomas Ellinor, Anthony Philippakis, Finale Doshi-Velez
In preparation to be submitted at ICML 2025.

Signature Activation: A Sparse Signal View for Holistic Saliency
Authors: **Jose Roberto Tello Ayala**, Akl C. Fahed, Weiwei Pan, Eugene V. Pomerantsev, Patrick Thomas Ellinor, Anthony Philippakis, Finale Doshi-Velez
Presented at ICML 3rd Workshop on Interpretable Machine Learning in Healthcare (IMLH). The paper was awarded the First Price for Best Paper at IMLH.

Optimal Testing and Containment Strategies for Universities in Mexico amid COVID-19
Authors: Luis Benavides-Vázquez, Héctor Alonso Guzmán-Gutiérrez, Jakob Jonnerby, Philip Lazos, Edwin Lock, Francisco J. Marmolejo-Cossío, Ninad Rajgopal, and **José Roberto Tello-Ayala**.
Presented at the inaugural Association for Computing Machinery (ACM) conference on Equity and Access in Algorithms, Mechanisms and Optimization (EAAMO) 2021.

RESEARCH & WORK EXPERIENCE

Business Intelligence Manager
Media Maker Group, based in Cuernavaca Morelos, Mexico.
I implemented machine learning, optimization, and data science solutions to improve Out-Of-Home advertisement selection utilizing geo-refencenced data.
From September of 2021 until June 30 of 2022.

Research Assistant
Test and Contain, a project originated at the University of Oxford and the San Luis Potosí Institute of Scientific Research and Technology in Mexico.
I worked on designing mechanisms for testing and safely opening Mexican universities during the COVID-19 pandemic, based on multi-objective optimization, reinforcement learning, and information theory approaches.
From June of 2021 to January of 2022.

Bachelor's Thesis
My dissertation "A study concerning the Chebyshev Center problem in finite-dimensional normed vector spaces" is about geometry, approximation theory, and optimization on Banach spaces, focusing on the Chebyshev Center problem and its applications and with an emphasis on finite-dimensional spaces.
Defended my thesis on December 9 of 2020.

AWARDS

• Eric and Wendy Schmidt Fellow for Fall 2024.

• First place for the best paper award for our paper **Signature Activation: A Sparse Signal View for Holistic Saliency** at ICML 3rd Workshop on Interpretable Machine Learning in Healthcare (IMLH) ($1,000).

• Jerry A. and Adi K. Greenberg Fellowship for the Academic Year 2022-2023 (2023).

•Special Mention for my undergraduate dissertation (2020).

• Obtained a 60 % scholarship for my Bachelor Studies at ITAM (2014).

PRESENTATIONS/TALKS

Poster presentation at The Cardiovascular Research Center Retreat 2023, Massachusetts General Hospital. Rockport Massachusetts, October 13, 2023. For our upcoming work *CorAI: Artificial Intelligence Based Detection of Coronary Artery Stenosis in more than 40,000 Coronary Angiograms.* Akl C. Fahed*, Jose Roberto Tello Ayala*, Samuel Friedman*, Ozan Unlu, Marcus D. R. Klarqvist, Paolo Di Achille, Mustafa Nasir-Moin, Eugene Pomerantsev, Mahnaz Maddah, Finale Doshi-Velez, Patrick T. Ellinor, Anthony Philippakis.

Oral presentation at the 3rd Workshop on Interpretable Machine Learning in Healthcare (IMLH) at the International Conference in Machine Learning. Honolulu Hawaii, July 28, 2023. Presenting our paper *Signature Activation: A Sparse Signal View for Holistic Saliency* Jose Roberto Tello Ayala, Akl C. Fahed, Weiwei Pan, Eugene V. Pomerantsev, Patrick Thomas Ellinor, Anthony Philippakis, Finale Doshi-Velez.

Oral Presentation at the EconCS Seminar of Harvard SEAS. September 10, 2021. Online Presenting our work *Optimal Testing and Containment Strategies for Universities in Mexico amid COVID-19* Luis Benavides-Vázquez, Héctor Alonso Guzmán-Gutiérrez, Jakob Jonnerby, Philip Lazos, Edwin Lock, Francisco J. Marmolejo-Cossío, Ninad Rajgopal, and José Roberto Tello-Ayala.

TEACHING EXPERIENCE

Teaching Fellow
Harvard's Computer Science Department
For the course "CS282r: Inverse Reinforcement Learning". Taught by Professor Finale Doshi-Velez.
Fall semester 2023.

Instructor

Department of Statistics at ITAM
For two iterations of the course "Introduction to the R programming language". Gave recitations for a total of nine hours for each course to fellow students.
During September of 2018 and January of 2019.

Teaching Assistant
Department of Mathematics at ITAM
For the course "Differential and Integral Calculus I" (MAT - 14100). Gave recitations for four hours each week.
During the Fall Semester of 2017 and Spring Semester of 2018.